Mauricio Angel Treviño IV

Houston, TX | Portfolio | linkedin.com/in/mauricio-trevino4/ | maurit2001@gmail.com | U.S. Citizen

EDUCATION

Rice University, Houston, TX — August 2025 – Present
Master of Science in Space Studies, Expected Graduation December 2026 — GPA: 4.0/4.0
Current Coursework: Computational Fluid Mechanics, Solar System Physics, Data Science for Engineering Leaders

Texas A&M University, College Station, TX — August 2020 – May 2025
Bachelor of Science in Mechanical Engineering, Engineering Honors, Summa Cum Laude — GPA: 3.9/4.0

TECHNICAL SKILLS

- **CAD:** SolidWorks, CREO, Fusion 360 | **FEA:** SolidWorks Simulation, ANSYS
- **CFD:** ANSYS Fluent | **Simulation & Analysis:** MATLAB, FloMaster, LabVIEW | **Programming:** Python
- **Manufacturing:** CNC Mill (MasterCAM), FDM 3D Printing, Manual Tools | **Circuit Design:** Multisim

PROFESSIONAL EXPERIENCE

Society of Automotive Engineers Formula Internal Combustion, College Station, TX — May 2024 – May 2025
Aerodynamics Sub-Team Member
- Tuned event-specific setups using simulation and track validation, contributing to 2nd place overall at the 2025 FSAE Michigan competition
- Led aerodynamic development through full-car CFD simulations in Ansys Fluent, optimizing interactions between wings, undertray, and bodywork for performance gains
- Directed cross-subteam collaboration between aerodynamics, suspension, powertrain, and chassis to ensure cohesive vehicle integration and maximize overall performance
- Manufactured the full aerodynamics package using carbon fiber epoxy composites

Robotics and Automation Design (RAD) Lab, College Station, TX — May 2024 – July 2025
Undergraduate Research Assistant
- Engineered a deployable sensor package launched from RoboBall III to extend mission range, provide GPS localization, and enable visual reconnaissance
- Developed and integrated key subsystems for RoboBall II, including a wireless inductive charging module, real-time data visualization via ROS2/MATLAB, and explored soft-shelled mobility platforms for diverse environments
- Co-authored 1 published work, viewable on my ORCiD page: 0009-0002-2986-3486

NASA Pathways Co-Op, Stennis Space Center, MS — January 2023 - August 2023
Rocket Propulsion Test Engineer Co-Op — January 2024 - May 2024
- Designed and analyzed cryogenic piping and instrumentation systems for rocket propulsion testing, adhering to ASME standards and NASA protocols
- Conducted liquid oxygen transfers and supported test stand integration using model-based engineering approaches
- Evaluated novel valve performance, determining flow coefficients and assessing system-level impacts
- Selected as NASA PAXC Stennis Representative; presented technical work at the National Intern Symposium

Society of Automotive Engineers Aerospace Design, College Station, TX — April 2021 – May 2024
Product Owner of Structures and Material Science Sub-Team
- Optimized the internal truss structure of an RC aircraft using FEA and hand calculations
- Constructed the aircraft using precision fabrication techniques including custom jig assemblies
- Led the Micro Class team to 1st place overall at the 2024 SAE Aero Design West Competition, applying Agile methods to streamline development and reduce downtime

CERTIFICATIONS & INTERESTS

Certifications: Americas Cutting Edge (ACE) Manufacturing Certificate, HAAS Basic Mill and Lathe Certificates, Solidworks CAD Design Associate Certificate
Interests: Extraterrestrial Exploration, Lander Systems, Propulsion, Spikeball, FPV Drones

Design and Validation of a Dual-LiDAR Module for Real-Time Slope Estimation on Pressurized Spherical Rovers Descending Lunar Craters. iSpaRo 2025

Novel Robotic Fleet for Sample Recovery in Lunar Craters: A Concept of Operations. IEEE TFR 2025